

"Whole-Body Motion Planning in Wheeled Quadrupedal Robots"

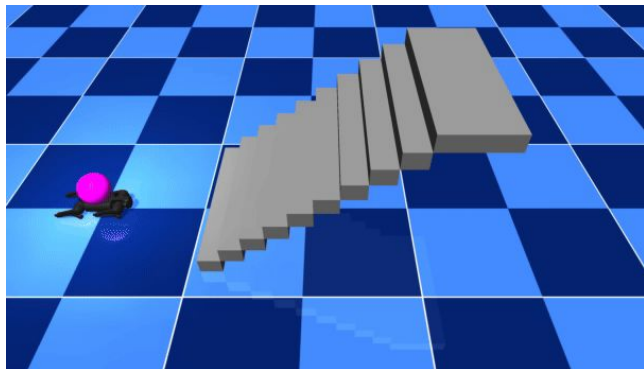
16-745 Optimal Control and Reinforcement Learning Project

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Motivation

Model Predictive Path Integral (MPPI) control has demonstrated effectiveness in generating real-time whole-body locomotion and manipulation behaviors for quadrupedal robots with legged locomotion. However, extending MPPI to wheeled quadrupedal robots presents new challenges due to **nonholonomic constraints, rolling dynamics, and hybrid locomotion requirements**.



Approach

1. MPX vs. iLQR
2. Incorporating rolling constraints
 - a. Project control samples onto constraint space before applying into sim
 - b. Or constrain dynamics model directly
 - c. Or sample unconstrained, then keep feasible control trajectories
3. Reformulating cost functions to penalize slippage and nonholonomic violations.
4. Optimizing wheel torques instead of joint-space torque control.
5. Stitching actions together (stretch)
6. Mujoco Physics Simulator

MPPI Method

Algorithm 1 MPPI Control Algorithm

Require: Initial state x_0 , control parameters (μ_0, Σ_0) , number of samples N , time horizon T , temperature parameter λ

Ensure: Optimal control input u

- 1: Initialize control trajectory mean μ_t and covariance Σ_t
 - 2: Initialize sample trajectory cost $\mathcal{L}_n = 0$
 - 3: **for** each sample $n = 1, \dots, N$ **do**
 - 4: **for** each timestep $t = 0, 1, \dots$ **do**
 - 5: Sample action sequence: $u_t \sim \mathcal{N}(\mu_t, \Sigma_t)$
 - 6: Simulate forward for T timesteps using u_n
 - 7: Compute the resulting cost and add it to $\mathcal{L}_n$
 - 8: **end for**
 - 9: **end for**
 - 10: Compute weights ω_n for each sample using Equation [1](#)
 - 11: Update control trajectory mean μ_t using Equation [2](#)
 - 12: Select the control input for the system: $u = \mu_t[0]$
 - 13: Apply control input u
 - 14: Shift control mean: $\mu_t \leftarrow \text{shifted}(\mu_t)$
-

The weight for the n -th trajectory is given by:

$$\omega_n = \frac{\exp\left(-\frac{\mathcal{L}_n - \mathcal{L}_{\min}}{\lambda}\right)}{\sum_{n=1}^N \exp\left(-\frac{\mathcal{L}_n - \mathcal{L}_{\min}}{\lambda}\right)}, \quad (1)$$

and the weighted control update is:

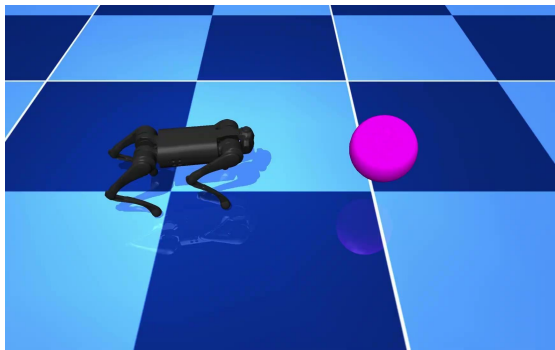
$$\mu_t = \sum_{n=1}^N \omega_n u_t, \quad (2)$$

where:

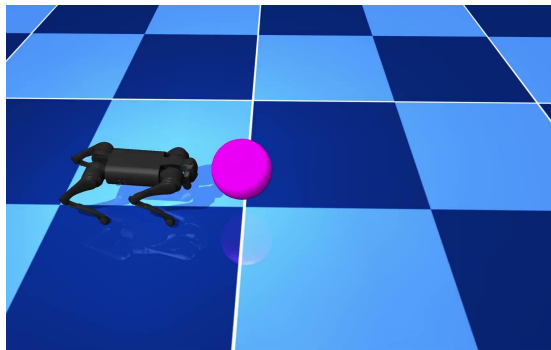
- $\mathcal{L}_n$ represents the **cost** of the n -th trajectory.
- $\mathcal{L}_{\min}$ is the **minimum cost** among all sampled trajectories.
- ω_n denotes the **normalized weight** assigned to the n -th trajectory.
- λ is the **temperature parameter** that determines the controller's sensitivity to differences in trajectory cost.
- u_t is the **control input** for the n -th trajectory.
- μ_t is the **expected control input**, computed as a weighted sum of sampled controls.

Preliminary Results: MPPI Action Rollouts

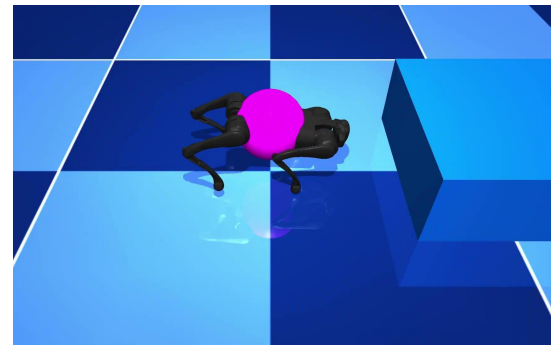
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Straight walking



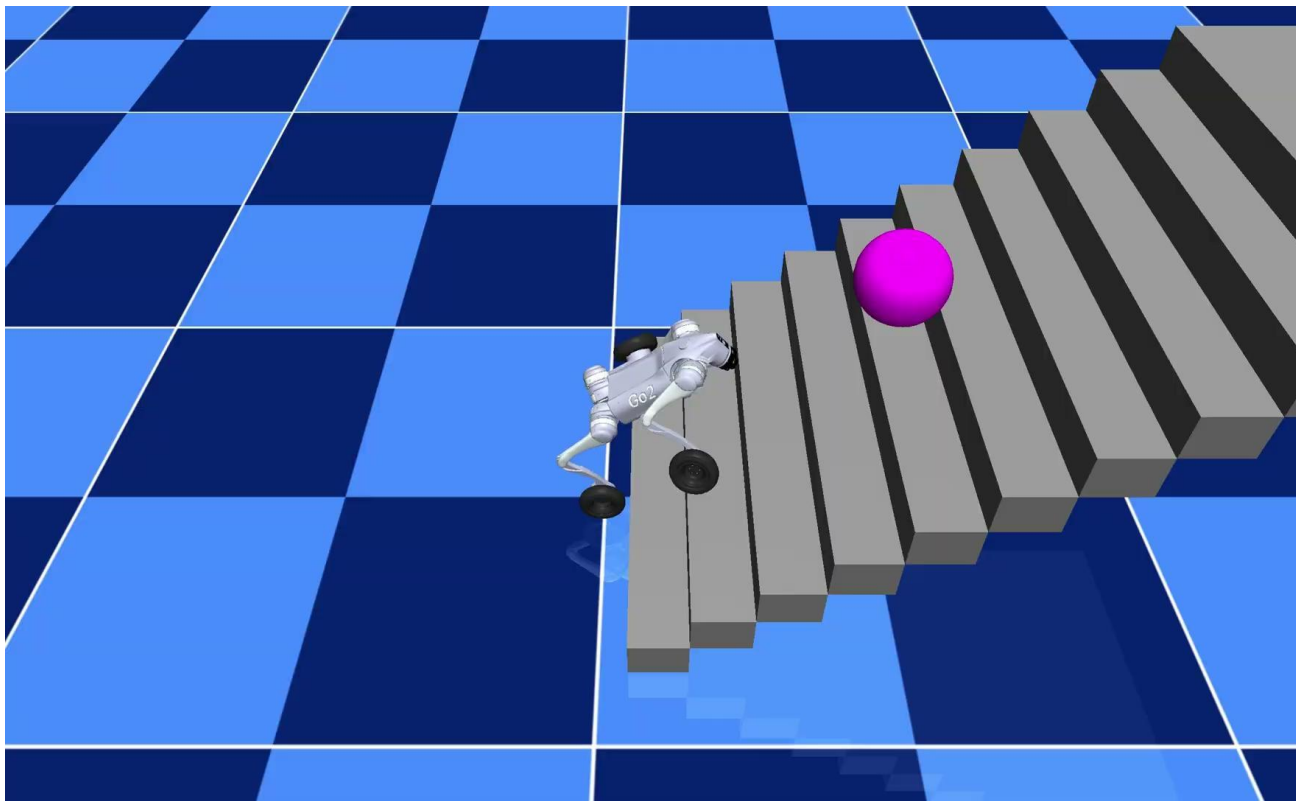
Waypoint walking



Box Jump

Wheeled Quad Results

Wheeled Stairs

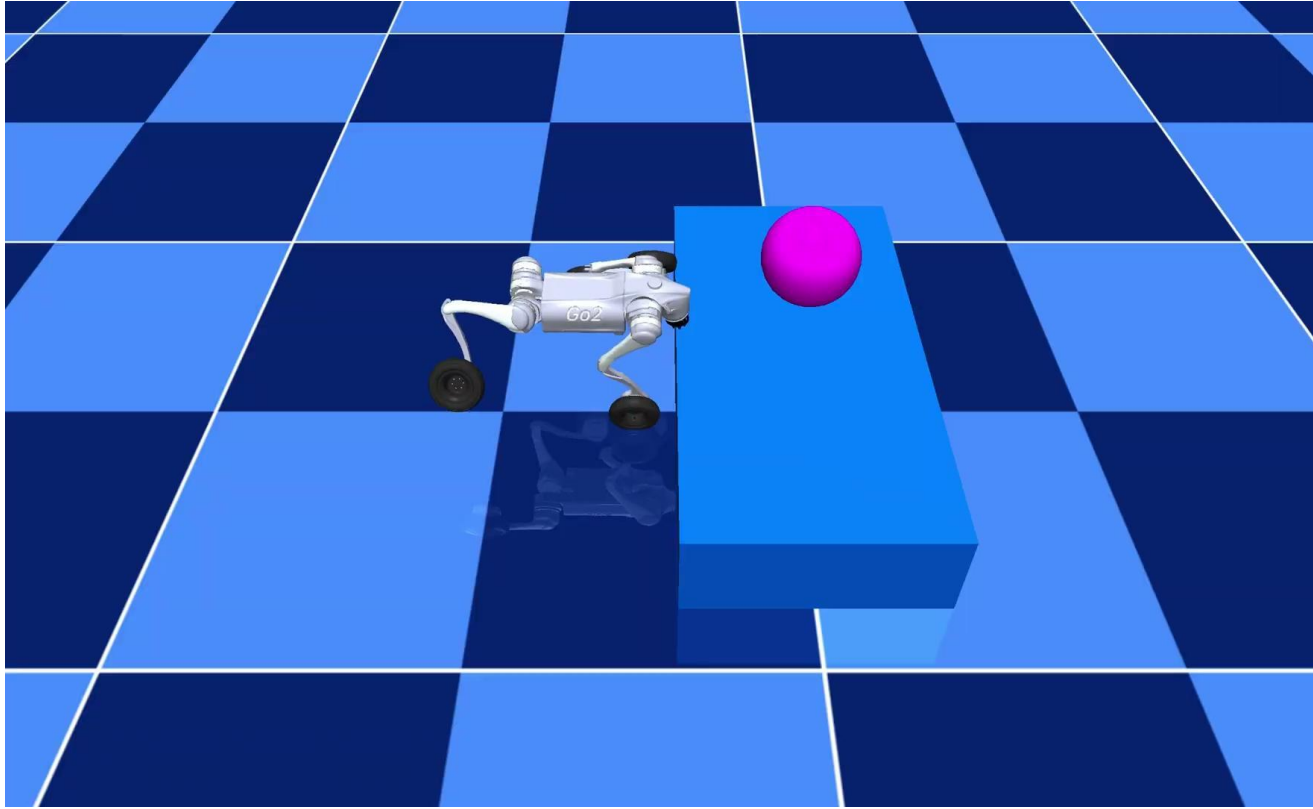


1x speed

Step height: 7.5 cm

Step length: 10 cm

Wheeled Box Jump

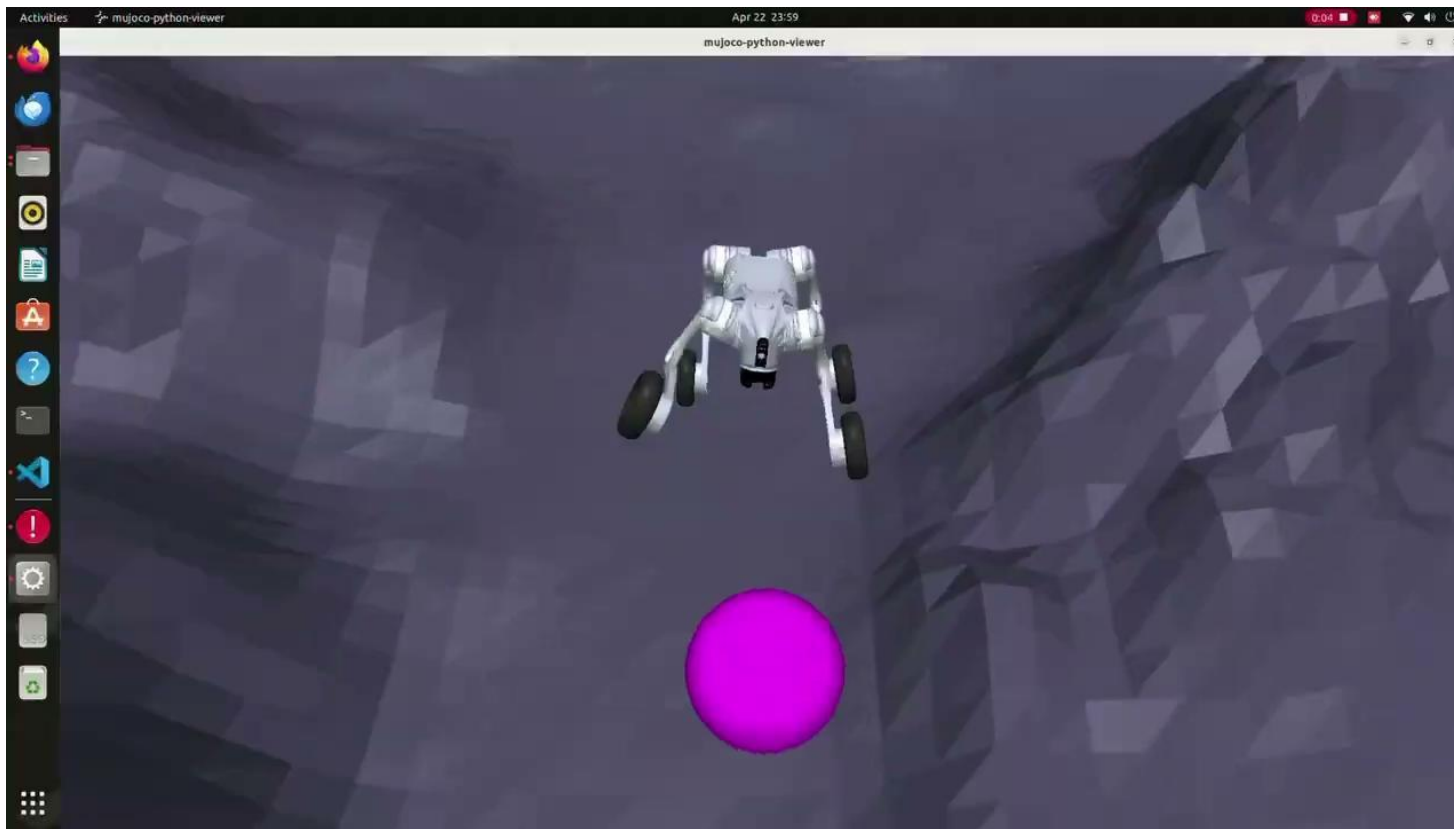


1x speed

Robot nominal
height: 40 cm

Box height: 30cm

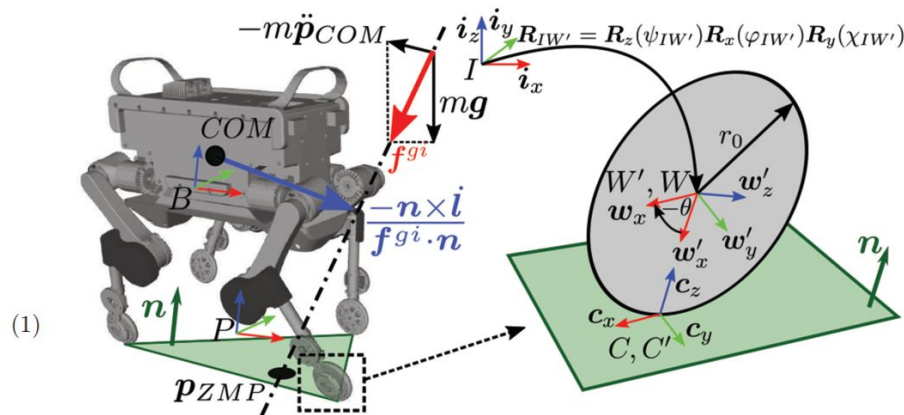
Rough Terrain - Sim demo



- Bjelonic describes acceleration of wheel contact point (Eq 1)
- Key features to capture:
 - Non-holonomic rolling constraint (no slip)
 - Rolling dynamics based on wheel parameters
 - Generalization to whole body of quadruped

$$\ddot{r}_{ICi} = J_{Ci}\dot{u} + \dot{J}_{Ci}u = R_{IW_i} \begin{bmatrix} 0 \\ -r_0\dot{\psi}_{IW'} \cos(\phi_{IW'}) (\dot{\chi}_{IW'} + \dot{\theta}) \\ r_0(\dot{\chi}_{IW'} + \dot{\theta})(\dot{\chi}_{IW'} + \dot{\theta} + \dot{\psi}_{IW'} \sin(\phi_{IW'})) \end{bmatrix}$$

- J_{Ci} is the Jacobian of the contact point,
- R_{IW_i} is the rotation matrix of the wheel frame relative to the inertial frame,
- r_0 is the wheel radius,
- θ is the wheel rotation angle.



References

- [1] R. Alvarez-Padilla, et al., "Real-Time Whole-Body Control of a Quadruped Using Model Predictive Path Integral Control," *IEEE Robotics and Automation Letters*, 2024.
- [2] M. Bjelonic, C. D. Bellicoso, Y. de Viragh, D. Sako, F. D. Tresoldi, F. Jenelten, and M. Hutter, "Keep Rollin'—Whole-Body Motion Control and Planning for Wheeled Quadrupedal Robots," *IEEE Robotics and Automation Letters*, vol. 4, no. 2, pp. 2116-2123, Apr. 2019.